

Upadesh Subedi

Computational Materials Science

Phase-Field Modeling | Multiphysics Simulation | Machine Learning | Nuclear Materials

upadesh.subedi@gmail.com | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#) | [ORCID](#) | subediupadesh.github.io

Postdoctoral Appointment

Incoming Postdoctoral Fellow, McMaster University, Canada

Jan 2027 – Jan 2029

Canada Impact+ Research Training Award | Modeling of Nuclear Fusion Reactor Materials

Education

Doctor of Philosophy in Materials Engineering

Oct 2022 – Sep 2026

Silesian University of Technology (SUT), Poland

Dissertation: *Towards Digital Twins for Quantifying Laser-Microstructure Interface in Multicomponent Alloys Using Thermodynamic Tensor Model*

Master of Science in Mechanical Design and Manufacturing

Apr 2019 – Sep 2021

Tribhuvan University (TU), Nepal

Bachelor of Engineering in Mechanical Engineering

Sep 2012 – Oct 2016

Tribhuvan University (TU), Nepal

Technical Expertise

Computational Modeling	Phase-Field, FEM, Multiphysics Simulations, MOOSE Framework
Machine Learning	Conv-LSTM, U-Net, Attention-based Neural Networks
Tensor Computing	Tensor Decomposition, Low-Rank Reconstruction, Data Inpainting
Programming	Python, C++ (MOOSE), Scientific WebApp/GUI development, Slurm HPC

Research Experience

Current Research

Computational Image Analysis of Irradiation-Induced Voids in High-Entropy Alloys

Collaboration: Prof. [Javier Dominguez](#), National Centre for Nuclear Research, Poland

- Developing a WebApp for automated STEM void segmentation and quantification
- Correlating void morphology with irradiation damage and depth

Damage Modeling with Grain Evolution in Nuclear Fusion Reactor Materials

Ongoing independent research

- Modeling polycrystalline grain evolution under magnetic and thermal fields
- Simulating orientation-dependent magnetic energy and tungsten intermetallic evolution

Doctoral Research

Oct 2022 – Sep 2026

Supervisors: Prof. [Tomasz Tanski](#) (SUT), Prof. [Anil Kunwar](#) (SUT), Prof. [Nele Moelans](#) (KU Leuven)

Multiphysics Modeling for Laser Processing in HEAs with Cross-Attention Transformers

- Coupled multiphysics and phase-field modeling of CoCrFeNi laser processing
- Cross-attention transformer for spatiotemporal field prediction and interpolation

AI-Based Virtual Digital Twins for Laser-Induced Phase Transition

- Modeled laser-driven phase evolution under diverse beam profiles
- Developed U-Net virtual digital twins ([AMTwinLab](#)) for real-time phase prediction

Phase-Field Framework for Laser Processing of Ti₃Au Intermetallics

- Modeled non-isothermal Ti–Au phase transformation during laser processing
- Quantified meltpool dynamics, IMC growth, and heat-mass diffusion lag

Spatiotemporal Neural Network Model for Alloy Interface Forecasting

- Developed Conv-LSTM for multigenerational microstructure evolution
- Demonstrated cross-material prediction for Pd–Rh and Cu–Mn alloys

Tensor-Based Reduced Representation of Microstructure Data

- Applied tensor decomposition to phase-field microstructure datasets
- Achieved 28× compression while preserving elastic deformation fidelity

Master's Research

Apr 2019 – Oct 2021

Supervisor: Prof. [Khem Gyanwali](#) (TU)

Machine Learning for Multi-Principal Element Alloys

- Developed neural-network phase prediction from multicomponent alloy compositions
- Open-sourced the [pyMPEALab](#) toolkit for alloy phase prediction and analysis

Peer-Reviewed Publications

1. M. Dagnaw, S. Poudel, **Upadesh Subedi**, et al.. [Integrating experiments and phase field method through informatics for tailored corrosion performance of additively manufactured steel microstructures](#). *Metals and Materials International*, Accepted, 2026.
2. **Upadesh Subedi**, et al.. [Spatiotemporally learning the morphological and dynamic features of Au meltpool under different heat source types with phase-field informed U-Net framework](#). *Computational Materials Science*, 2026.
3. **Upadesh Subedi**, et al.. [A non-isothermal multi-phase field approach to model the meltpool and IMC grains interaction in Ti–Au material](#). *Computational Materials Science*, 2025.
4. **Upadesh Subedi**, et al.. [Foretelling Microstructural Interface with Multi-Generational Convolutional-LSTM Framework](#). *Engineering with Computers*, 2025.
5. Hana Mamo, **Upadesh Subedi**, et al.. [Prototyping Ti₂Cu intermetallic grain growth heterogeneously in Ti6Al4V matrix through laser additive manufacturing](#). *Materials & Design*, 2024.
6. **Upadesh Subedi**, et al.. [Rapid portabilization of elasto-chemical evolution data for dental Ti–Cr alloy microstructure through sparsification and tensor computation](#). *Scripta Materialia*, 2024.
7. Sachin Poudel, **Upadesh Subedi**, et al.. [AlloyManufacturingNet for discovery and design of hardness-elongation synergy in multi-principal element alloys](#). *Engineering Applications of Artificial Intelligence*, 2024.
8. S. Liang, C. Wei, A. Kunwar, **Upadesh Subedi**, et al.. [Phase field modelling combined with data-driven approach to unravel the orientation influenced growth of interfacial Cu₆Sn₅ intermetallics under electric current stressing](#). *Surfaces and Interfaces*, 2023.
9. **Upadesh Subedi**, et al.. [State-of-the-Art Review on the Aspects of Martensitic Alloys Studied via Machine Learning](#). *Metals*, 2022.
10. **Upadesh Subedi**, et al.. [Automatic Featurization Aided Data-Driven Method for Estimating the Presence of Intermetallic Phase in Multi-Principal Element Alloys](#). *Metals*, 2022.
11. **Upadesh Subedi**, et al.. [pyMPEALab Toolkit for Accelerating Phase Design in Multi-principal Element Alloys](#). *Metals and Materials International*, 2022.

Research Software Developed

pyMPEALab	Open-source GUI toolkit for MPEA phase prediction and analysis
IMCATHEA	Neural-network WebApp for intermetallic phase prediction in HEAs/MPEAs
AMTwinLab	U-Net virtual digital twin for laser-induced phase evolution
VoidScope	Automated STEM void segmentation, morphology, and depth quantification

Academic Employment

Research Assistant, Silesian University of Technology, Poland

Jan 2023 – Jun 2026

Research team member in projects funded by [National Science Centre](#), Poland

- Led multiphysics and machine-learning development for materials modeling
- Co-authored eight peer-reviewed publications, including four as first author

Research Assistant, Tribhuvan University, Nepal

Jan 2020 – Sep 2022

Energy Systems Research Laboratory

- Developed machine-learning models for MPEA phase prediction
- Published three first-author papers and open-sourced [pyMPEALab](#) and [IMCATHEA](#)

Visiting Scholar

Department of Materials Engineering, KU Leuven, Belgium

Aug 2023 – Sep 2023

Supervisor: Prof. [Nele Moelans](#)

- Phase-field microstructure simulation using Vlaams Supercomputer Centrum (VSC/HPC)

Research Funding and Awards

- **Canada Impact+ Research Training Award**, Postdoctoral Fellowship, 2027–2029 (CAD 140,000)
- **Departmental Research Grants for Young Scientists**, SUT, 2023–2026 (EUR 8k)
- **Best PhD Student Award 2022/2023**, SUT (EUR 8.5k; International Mobility Funding)
- **Ranked 1st in M.Sc. cohort**, Institute of Engineering, TU, 2021

Teaching Experience

Teaching Assistant, Institute of Engineering, Tribhuvan University, Nepal

Jun 2021 – Jun 2022

Operations Research and Management Science for Undergraduates

- Delivered lectures, tutorials, and coding exercises in operations research
- Taught optimization using Excel Solver, Crystal Ball, and simulation

Academic and Editorial Service

Guest Editor, *Journal of Visualized Experiments (JoVE)*

Method Collection: [Computational Methods in Additive Manufacturing for Biomedical Applications](#)

Peer Reviewer, 15+ journal reviews

Nature Communications, npj Computational Materials, Computational Materials Science, Scientific Reports, Archives of Computational Methods in Engineering, Applied Physics A, and others.

Selected Conference Presentations

1. **Upadesh Subedi** (Presenting Author), et al. [Attention-guided surrogate reconstruction of multicomponent phase-field simulations for laser-processed CoCrFeNi](#). Physical Metallurgy & Materials Science Conference (AMT 2026), Poland.
2. **Upadesh Subedi** (Presenting Author), et al. [Digital Twin for Phase Transition Prediction in Laser-Based Additive Manufacturing: A Phase-Field and U-Net Model Framework](#). 5th Additive Manufacturing Meeting (AMM 2025), Poland.
3. **Upadesh Subedi** (Presenting Author), et al. [Temperature dependence of free energies in a phase field model for selective laser melting of tin powder on copper substrate](#). 5th International Symposium on Phase-Field Modelling in Materials Science (PF24), Zhejiang University, China.
4. A. Kunwar, **Upadesh Subedi**, et al. [Phase field modeling of thermomigration at colder interface of solid state solder joints](#). 1st International Conference on Mechanics of Solids, 2022.
5. A. Kunwar, J. Hektor, **Upadesh Subedi**, et al. [Mesoscale Simulations Guiding Data-driven Design of Electronic Materials](#). TMS 2022 Annual Meeting & Exhibition.